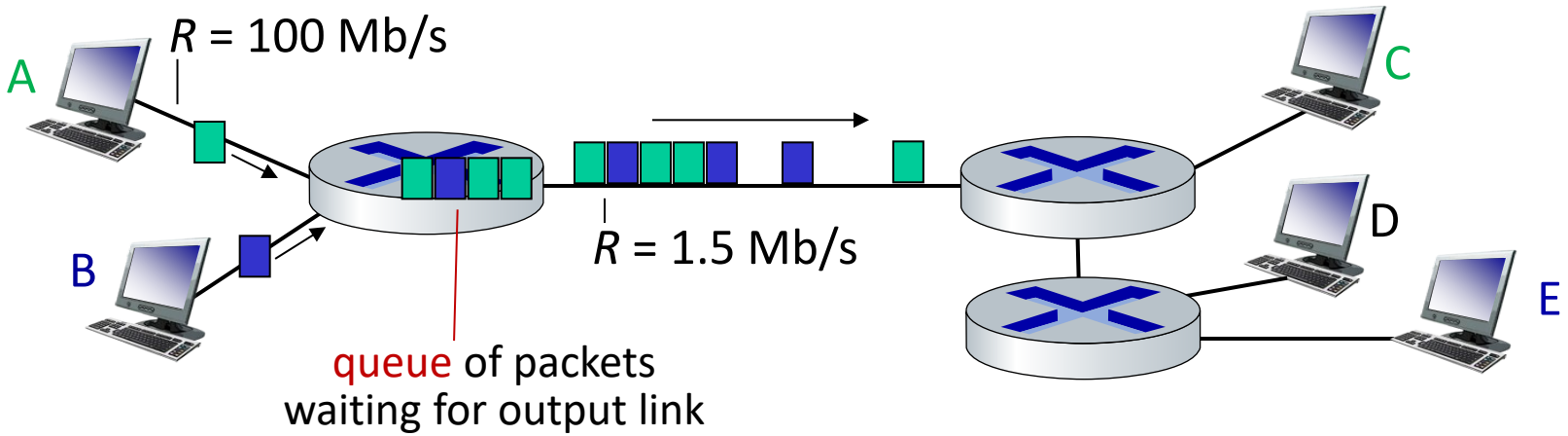
Lecture 5-Part 1

- Packet Loss and Throughput
- Network Layering

How do packet loss occur?



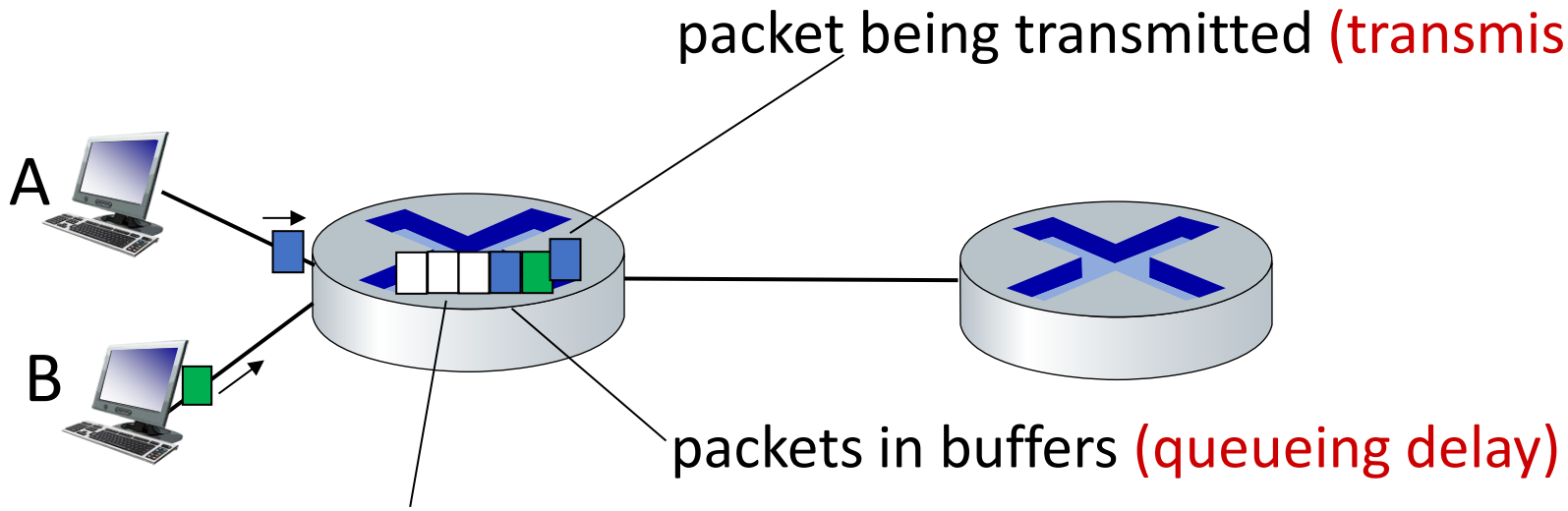
Packet queuing and loss: if arrival rate (in bps) exceeds transmission rate (bps) for a period of time

Demo: Link

How do packet loss occur?

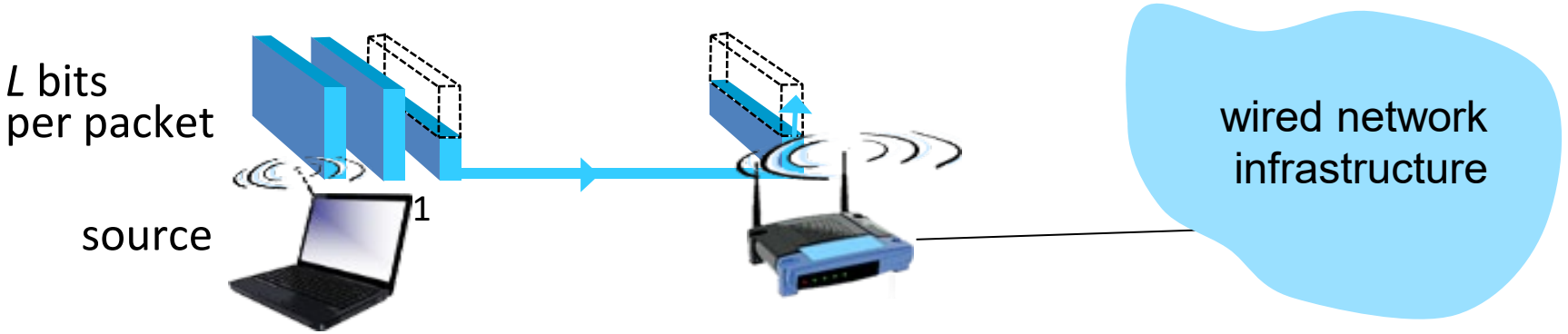
packets *queue* in router buffers

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up



How do packet loss occur?

- ***Other causes***: Packet loss could also happen due to channel errors, weak signals, etc. ➔ mostly over wireless links

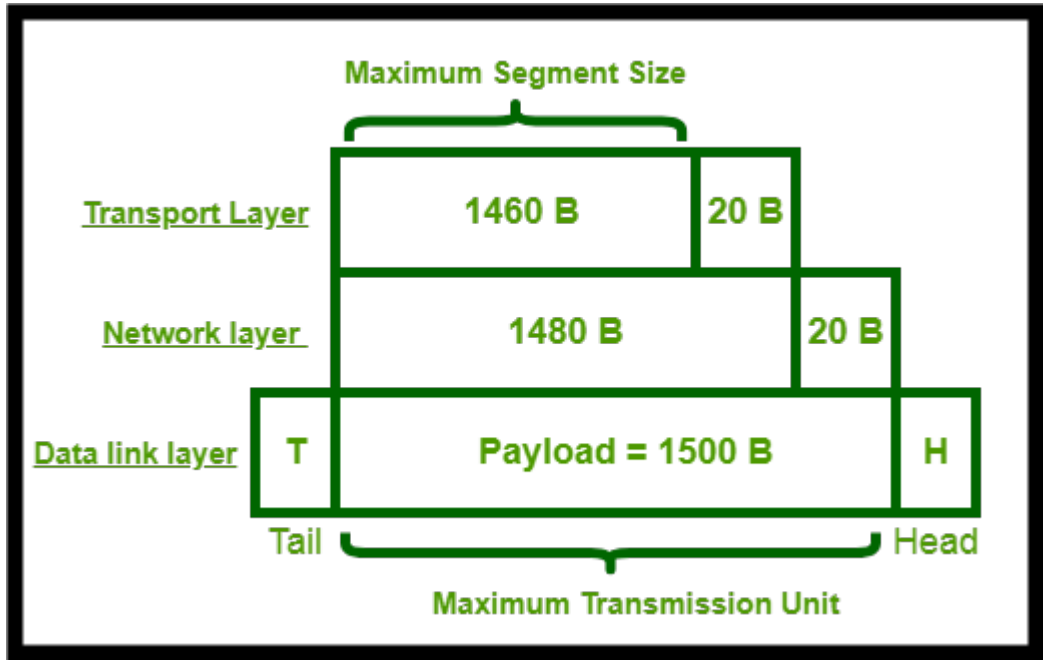


- Assume that each bit could be received in error with probability *p*.
- Probability that an **L-bit packet is received successfully**?
- Probability of packet error?

“Optimal” Packet Size in Packet-Switched Networks

- Any packet carries some data bits + header bits
  - *M* bits data (message)
  - *H* bits header
- There is a tradeoff in choosing the value of *M*
- For a fixed header size H
  - If *M* is too small
    - Ratio of message bits to header bits decreases

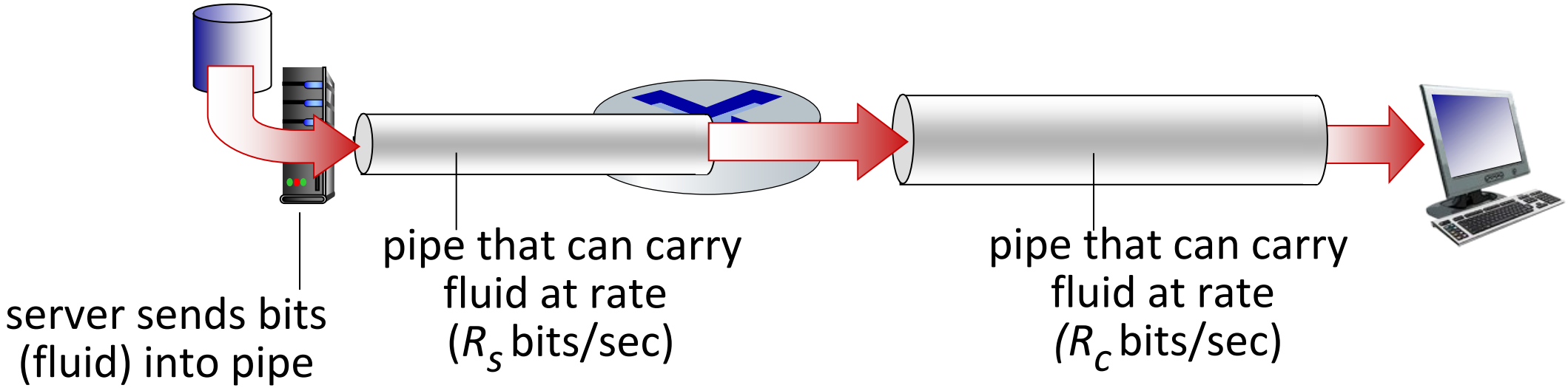
    ➔ Not efficient
  - If *M* is too large
    1. probability of packet error due to bit errors increases.
    2. excessive delay due to store-and-forward operation



Throughput

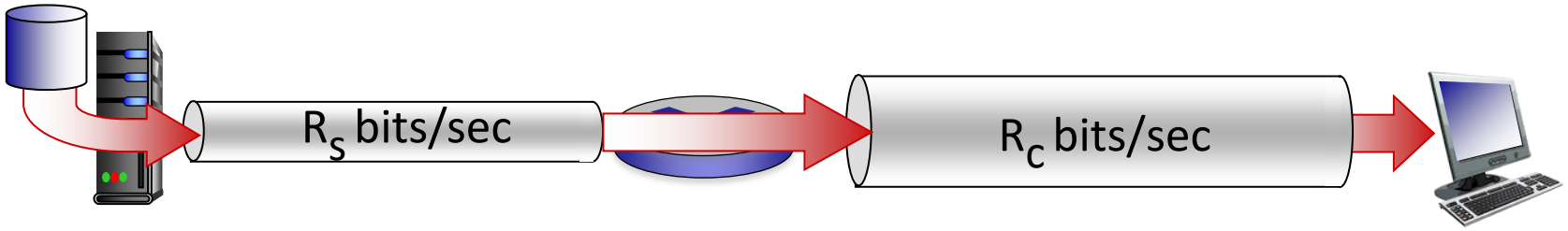
- *Throughput:* Transfer_size / Time_to_transfer ➔ Unit: Bit/Sec
  - Number of packets (bits, bytes, …) delivered per unit time, typically averaged over the duration of a connection
  - (Effective) bandwidth perceived by the user
  - *Instantaneous throughput:* rate at given point in time
  - *Average throughput:* rate over longer period of time

Throughput

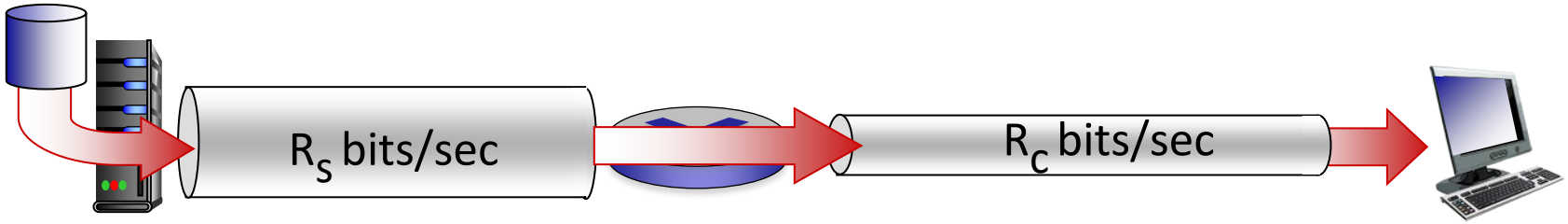

server sends bits
(fluid) into pipe
pipe that can carry
fluid at rate
(R_s bits/sec)
pipe that can carry
fluid at rate
(R_c bits/sec)

Throughput

$R_s < R_c$ What is average end-end throughput?



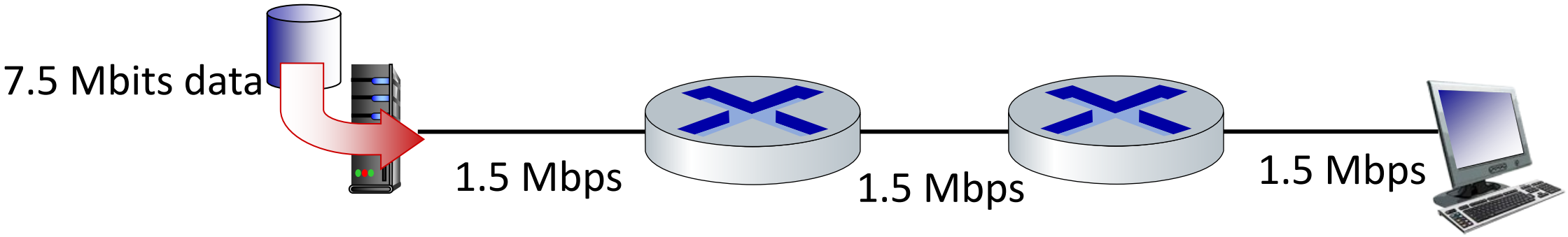
$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

End-End Throughput

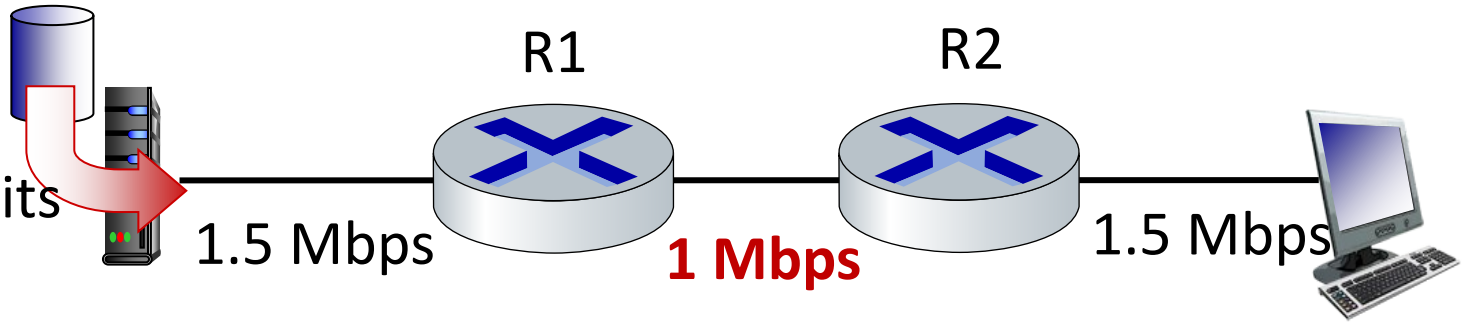


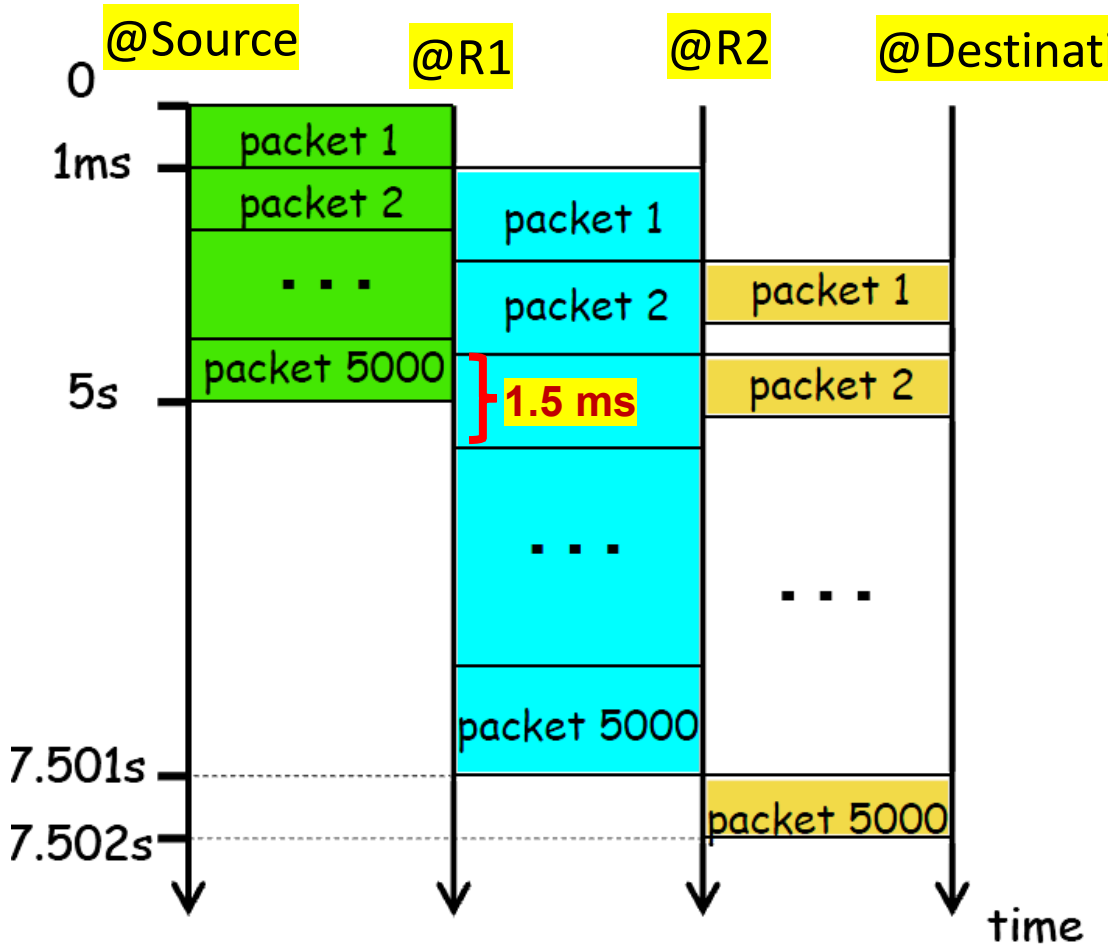
Example:

- Total amount of data = 7.5 Mbits
- Packet size L = 1,500 bits
- Link rate R = 1.5 Mbps
- End-End throughput = 1.5 Mbps

7.5 Mbits data

Packet Size: 1,500 bits





- Throughput: 7.5Mb/7.502s ~ 1Mbps
- Slowest (bottleneck) link dominates

Chapter 1: roadmap

- Basics of Network Building Blocks and Protocols
- Internet Structure
- Network Core: Packet vs. Circuit Switching
- Network Performance: Delay, Packet loss, Throughput
- Protocol Layers and Service Models

Protocol “layers” and reference models

Networks are complex, with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

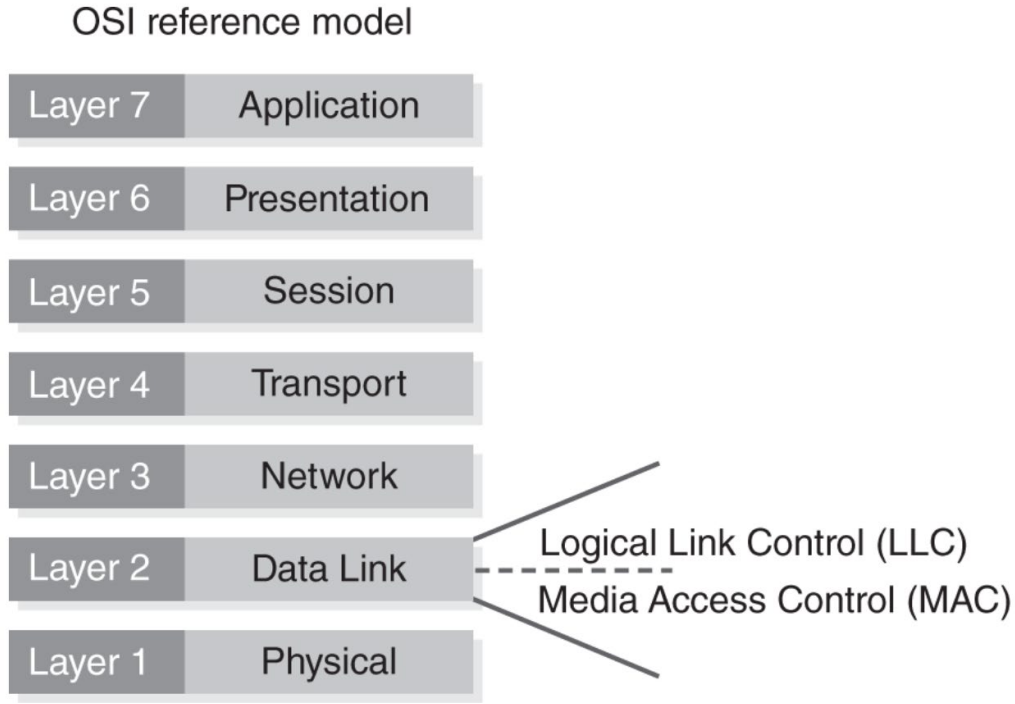
Question:

is there any hope of *organizing* structure of network?

…. or at least our *discussion* of networks?

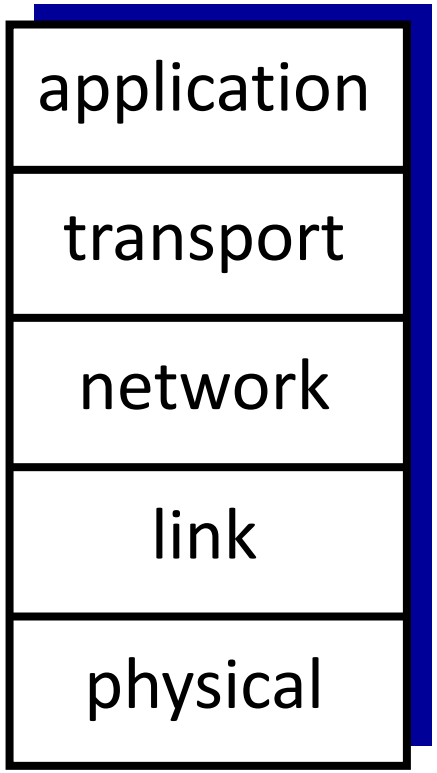
OSI Model

- Open Systems Interconnection (OSI) reference model
- Design technology for each layer without worrying about how other layers work
- Each layer need to know how to talk to the layer above and the layer below it
- Modularization: easer maintenance and updating
  - change in layer's service *implementation*: transparent to rest of system
- layering considered harmful?

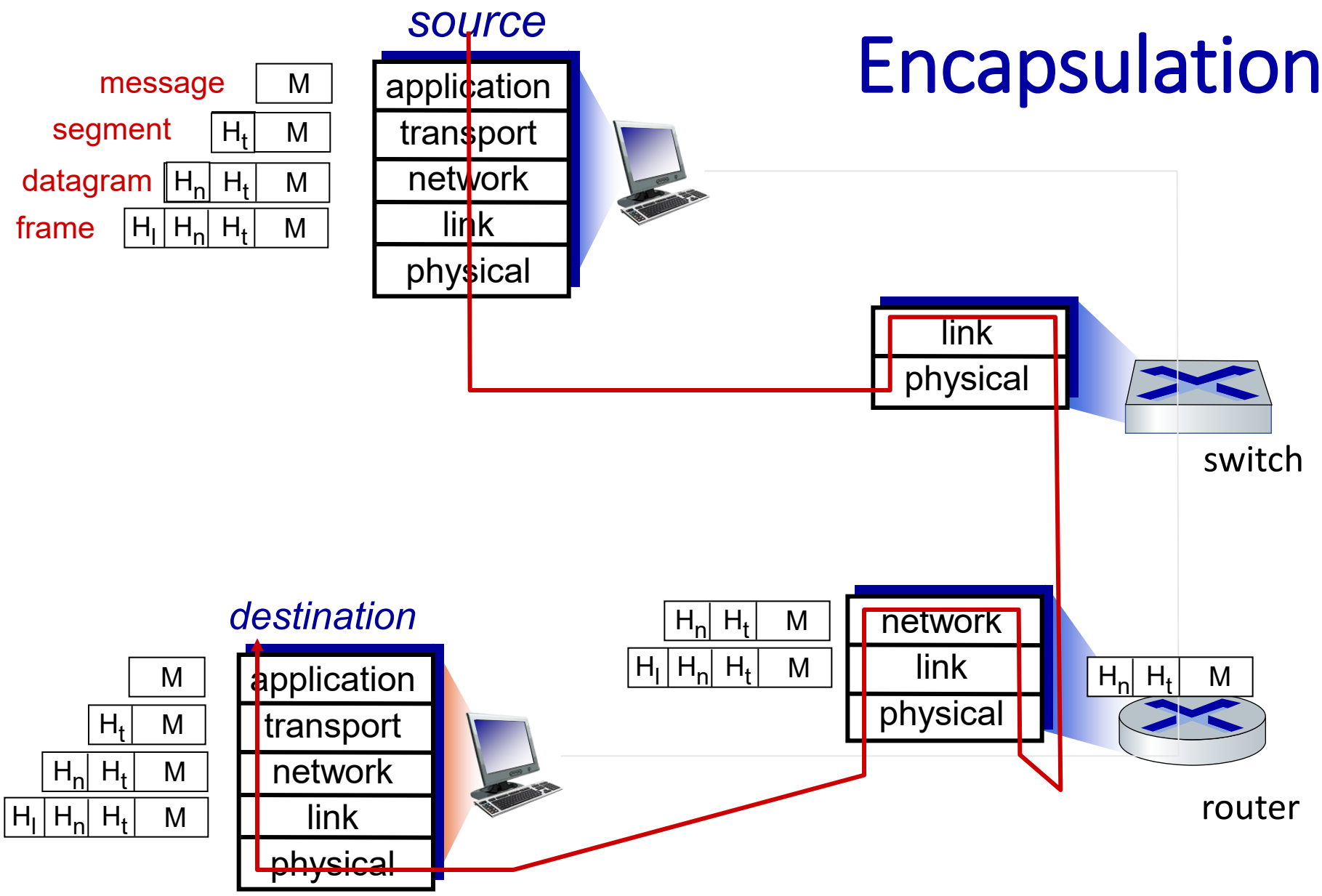


Simplified OSI Model: Internet protocol stack

- *application:* supporting network applications
  - ***Example protocols:*** IMAP, SMTP, HTTP
- *transport:* process-process data transfer
  - ***Example protocols:*** TCP, UDP
- *network:* routing of datagrams from source to destination
  - ***Example protocols:*** IP, routing protocols
- *link:* data transfer between neighboring network elements
  - ***Example protocols:*** Ethernet, 802.11 (WiFi)
- *physical:* bits “on the wire”



Encapsulation



Chapter 1: summary

Q1: What did we learn in this chapter?

Q2: What is the biggest question remaining?

Chapter 1: summary

We've covered the foundations of networking:

- Internet overview and structure
- Network edge and access network
- Network Core: packet-switching vs. circuit-switching
- Network resources and performance metrics: loss, delay, throughput
- Network layers

We now have:

- The context, overview, vocabulary, and "feel" of networking
- More depth and details to follow.